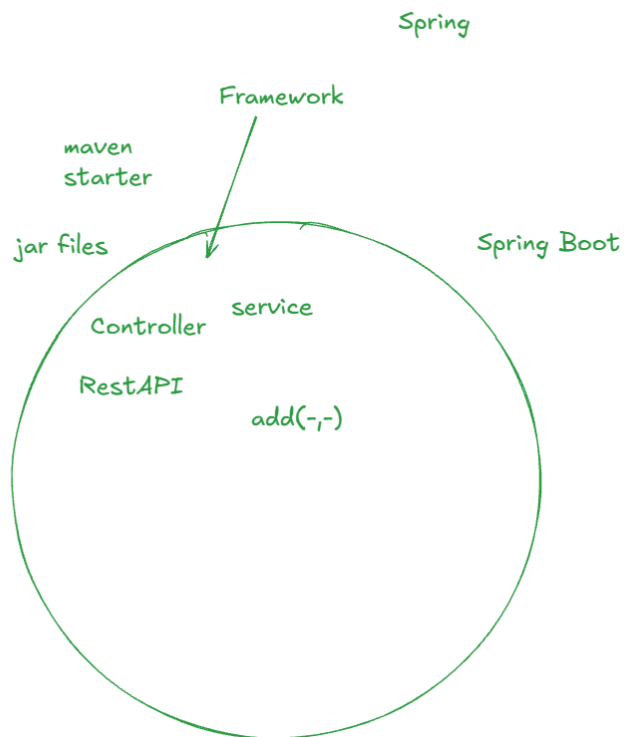
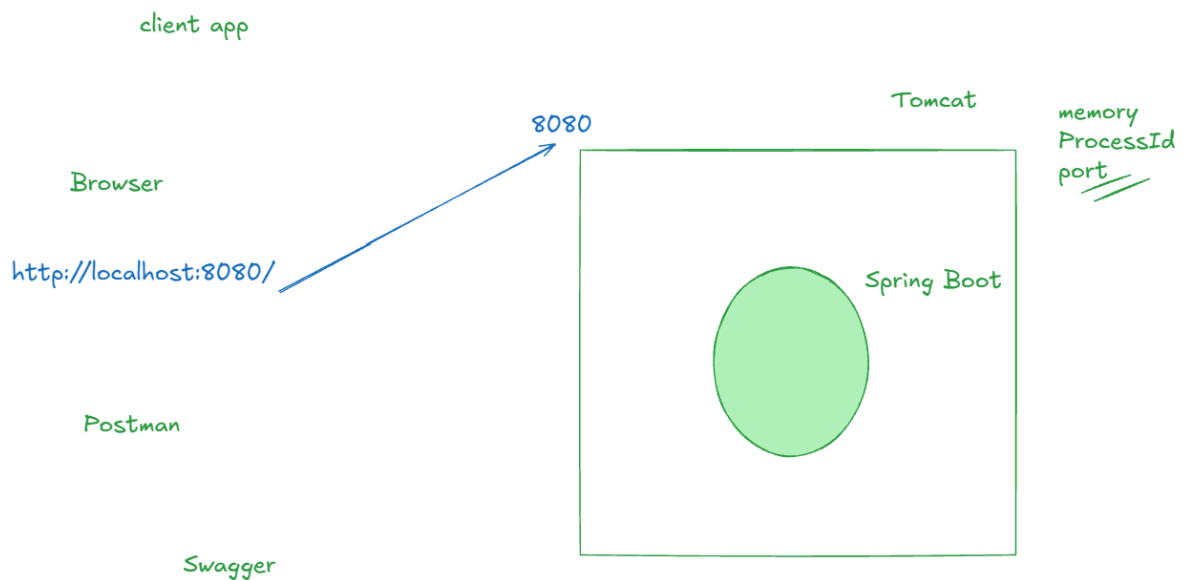


LIVE NOTES

Overview of using spring framework code along with your app to build a restapi



App runs in tomcat and exposes port 8080, clients can connect on that port.



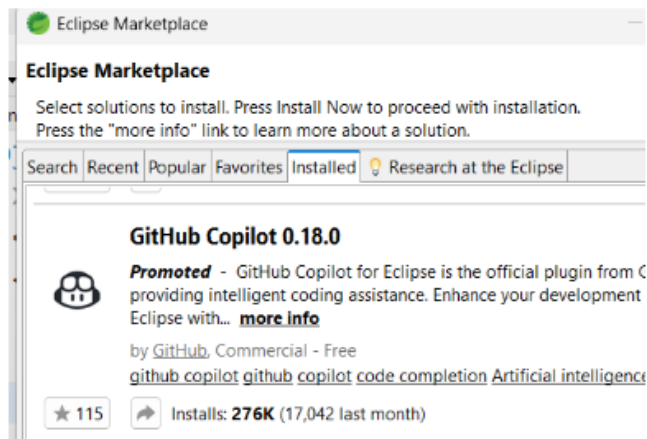
AI coding agent support

Application Development

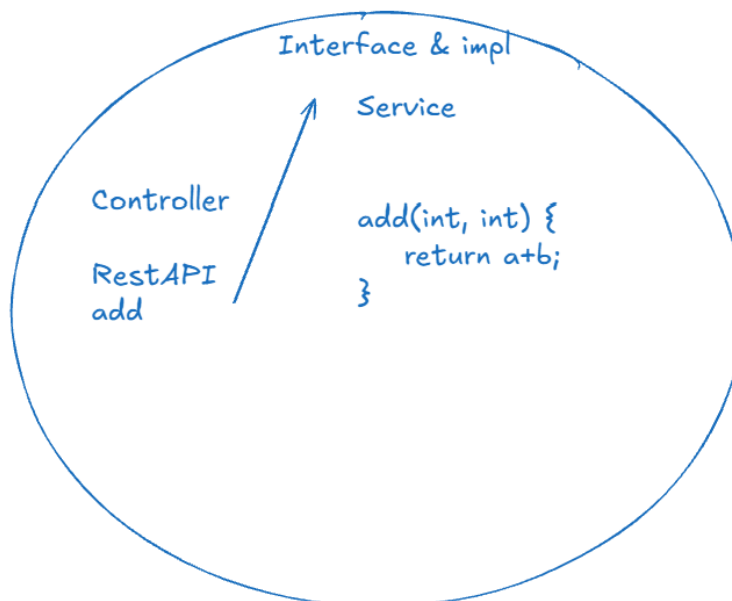
Github Copilot
ClaudeCode
Codex
Cursor
..
..

LLM

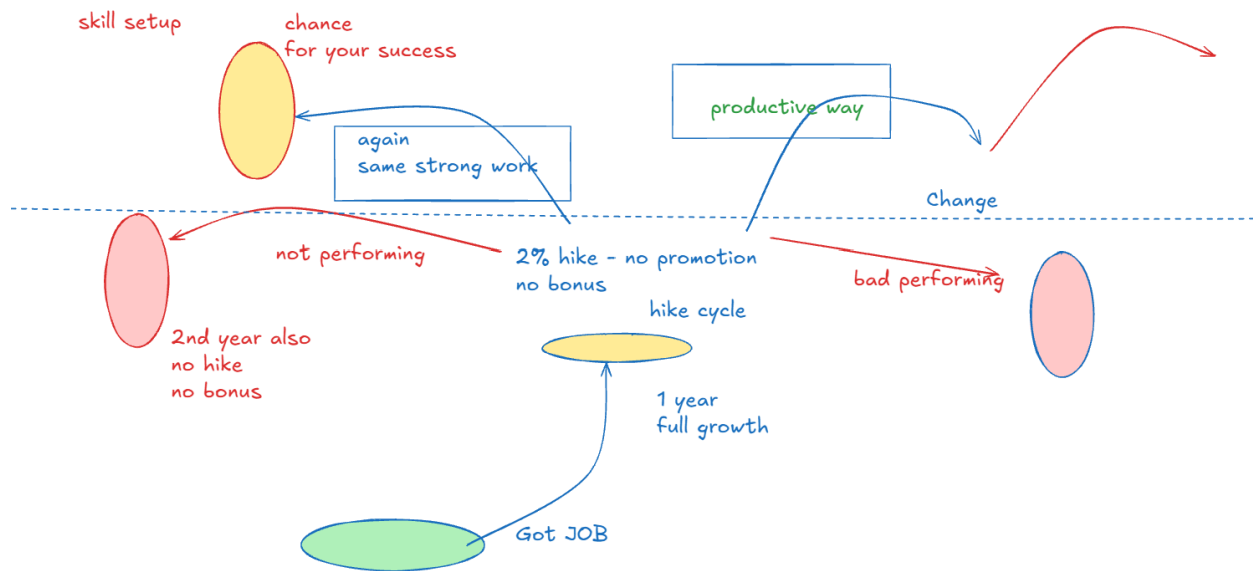
Add Github Copilot from eclipse market place



Controller talks to service based on interface



All throughout your life, you work for yourself.



Make a rest api

outside people can call our functionality

Framework

spring

project & modules -

GitHub

got the project into IDE.

maven commands to operate with this project.

Spring Boot application. - Rapid application development

Code the application

build - compile, prepare jar (artifact)

mvn clean package

=> GOAL

clean package

execute

mvn clean spring-boot:run

Update IDE setting to java 17.

C:\Program Files\Java\jdk-17

installed JRE => 17

compiler compliance level => 17

starting a "spring web" application

server (Tomcat)

Tomcat Default Embedded server given along with Spring Boot application.

java A

Tomcat - Default Embedded server needed to run our Spring Web application.

tomcat for your spring boot application, will run on default port 8080

invoker system will need to connect on 8080 to talk to this spring boot application.

=====

Browser

http://<where is tomcat running>:<what port>

http://localhost:8080

Type below url in browser.

http://localhost:8080

in localhost, is there any software running on port 8080 which can server http request

=====

Add capability to our IDE & Spring Boot application....

1. GitHub copilot

AI Coding Agents / AI coding assistant

ClaudeCode

Codex

Cursor

It talks to LLMs

1. Setup account

<https://github.com/features/copilot/plans>

2. in IDE eclipse market place

GitHub copilot

3. restart IDE (stop & relaunch)

4. Signin to your GitHub copilot account in IDE.

- Inline code suggestion

- Copilot Agent Mode

=====

Create a .controller.MathController, .service.MathService & MathServiceImpl

at service later create method add(int, int) and return sum value as int.

Make MathController as a RestAPI which can call service layer with 2 int values and returns the

response.

we want to invoke this add method from browser.

=====

call from browser

<http://localhost:8080/math/add?a=3&b=5>

use GitHub copilot to code controller & service

restart your spring boot app

test from browser

GitHub copilot

Agentic behaviour

- when we give a prompt
- Agent will perseve the environment.
- Baed on our prompt, make decision, of what to do.
- Start making changes accordingly
- during changes, if any mistakes happen, then automatically understand and fix those mistakes
- run it till entire task is accomplished.

2. Devtool

we want whenever we make changes to application, automatically the spring boot app should restart.

add maven devtool dependency in pom.xml

```
<dependencies>  
</dependencies>
```

devtool is a spring boot feature

try to remove version in dependency. it will automatically pick springboot version.
if not happening and leads to error, then you need to add the corresponding version number.

when you added changes to pom.xml
clean package
start your application.

3. Lombok

removing boiler plate code.

some common code, needed for the project, but it should happen automatically, we don't want to write it by ourselves

```
MyClass {  
    int v1;  
    int v2;
```

```
    //getter  
    //setter  
    //toString
```

exploring solutions by working with internet.
Use AI for problem solving & thinking
Brainstorm with AI.

IDE level

<https://projectlombok.org/download>

```
java -jar lombok.jar
```

Select your IDE location:

- It will open installer → detect your STS/Eclipse

Install / Update

Restart STS

Annotation processing

pom.xml

```
<dependency>
    <groupId>org.projectlombok</groupId>
    <artifactId>lombok</artifactId>
    <scope>provided</scope>
</dependency>
```

Test approach:

1. all saved
2. no compilation problem
3. no service startup problem
4. browser/postman for further testing

4. Logging

lot of coding is happening using AI

prompt

logging is mandatory for project
without effective logging we cannot deploy project into production.

logs need to tell the execution flow.
what values were taken at runtime
if there are any error, that should be highlighted.

don't write `System.out.println()`
on an object you call method

Slf4j with logback - default provided by spring boot

```
Logger logger
logger.info(---)
```

log4j, log4j2


```
log.info("Received request to add {} and {}", a, b);
```

```
[2m2026-06-10T12:10:15.865+05:30 [0;39m [32m INFO [0;39m [35m21116 [0;39m [2m---  
[demo] [nio-8080-exec-1] [0;39m [36m mc.mycomp.demo.controller.MathController [0;39m [2m:  
[0;39m Received request to add 3 and 500
```

```
@Slf4j  
public class MathController {  
  
    use log.info(-)  
}
```

how do you do logging into your project?

Working with AWS.

```
=====
```

deploy this spring boot application in AWS and test it there.

```
=====
```

1hr - 1.5h

Linux

mobaxterm

AWS EC2 machine deploying spring boot application

45 mins.

```
=====
```

13 Career success rules - needed for getting Job & for making successful career.

```
=====
```

```
=====
```

- Becoming an A Player! A, B, C, D.
become a A or B player.

A - Top people - 1/2%
if they are in system
projects grow
people grow
clients grow
tech company grows
pillar of the system
leaders of the system
they are ok to get no financial results, but still perform great
they are ok to sacrifice their own success, so the team becomes successful.

B - Exceed expectation
you are hired for 5lakh
job of 8lakh

forced to pay you around 8lakh.

DO MORE THAN YOU ARE PAID FOR.

C - Meets expectation
don't expect hike.
0 hike
ok bonus

inflation adjustment - 3%-5%-7%

D - Below expectation
5lakh
not able to do job
3lakh
0 hike
0 bonus
self improvement
terminated

HIRE FASTER, FIRE FASTER, PROMOTE FASTEST

=====

2. Work - Life success. 24hrs

Its not about Work-life balance its about work-life success.

Mon-Fri, 10AM - 7PM - work
everything else is life.

remove time from the equation.

work success

- great tech skills
- always having job - no fear of job
- financially successfully
- you can solve any tech problem
- you are constantly promoted
- you are respected at work
- you are valued
- your opnions matter
- your name is enough, if you are in team, that team will succeed
- you can lean anything new, every quickly
- you go to any interview and you will crack it
- everybody wants you in their team, & company
- the market needs you.

life success

- happiness
- joy
- art
- music
- fun
- friend
- family
- health
- travel
- latest luxury
- doing something better for the plannet
- peace of mind
- value in the life of other peole

- ,, what your life success looks like.

=====

3. All throughout your life you work only for yourself. Be Self Sufficient.

=====

Day4:

AWS deployment
Project
Continuing into Week2.
google meet

all the members who are enrolling for week2.
trainee offer letter will be issued this weekend.

Monday we will start with Week2 with the enrolled members.

Day1 - end

=====

Spring boot application.